

SIM1002 Calculus I
Tutorial 10
The Definite Integral

Exercise 5.4

Evaluating Integrals

Evaluating the integrals in Exercises 1-5.

1. $\int_{-1}^1 x^{299} dx$
2. $\int_0^{\frac{\pi}{3}} 4 \left(\frac{\sin u}{\cos^2 u} \right) du$
3. $\int_0^{\frac{\pi}{6}} (\sec x + \tan x)^2 dx$
4. $\int_1^8 \frac{\left(\frac{1}{x^{\frac{1}{3}} + 1} \right) \left(2 - x^{\frac{2}{3}} \right)}{x^{\frac{1}{3}}} dx$
5. $\int_0^{\pi} \frac{1}{2} (\cos x + |\cos x|) dx$

In Exercises 6-7, guess an antiderivative for the integrand function. Evaluate the given definite integral.

6. $\int_1^{\pi^2} \left(\frac{\sin(\sqrt{x})}{\sqrt{x}} \right) dx$
7. $\int_0^{\frac{\pi}{3}} \sin^2 x \cos x dx$

Derivatives of Integrals

Find $\frac{dy}{dx}$ in Exercises 8-12.

8. $y = \int_1^{\sin x} 3t^2 dt$
9. $y = \int_0^{\sqrt{t}} (x^4 + x^2) dx$
10. $y = x \int_2^{x^2} \sin(t^3) dt$
11. $y = \left(\int_0^x (t^3 + 1)^{10} dt \right)^3$
12. $y = \int_{\tan x}^0 \frac{dt}{1+t^2}$

Exercises 5.5

In Exercises 1-4, make the given substitutions to evaluate the indefinite integrals.

1. $\int \frac{(1+\sqrt{x})^{\frac{1}{3}}}{\sqrt{x}} dx, u = 1 + \sqrt{x}$

2. $\int \left(1 - \cos\left(\frac{t}{2}\right)\right)^2 \sin\left(\frac{t}{2}\right) dt, u = 1 - \cos\left(\frac{t}{2}\right)$

3. $\int \frac{1}{x^2} \cos^2\left(\frac{1}{x}\right) dx, u = \frac{1}{x}$

4. $\int \frac{dx}{\sqrt{5x+8}}$

(a) Using $u = 5x + 8$

(b) Using $u = \sqrt{5x + 8}$

Evaluate the integrals in Exercises 5-10.

5. $\int \sqrt{\sin x} \cos^3 x dx$

6. $\int \operatorname{cosec}\left(\frac{v-\pi}{2}\right) \cot\left(\frac{v-\pi}{2}\right) dv$

7. $\int \frac{\cos(\sqrt{\theta})}{\sqrt{\theta} \sin^2(\sqrt{\theta})} d\theta$

8. $\int \frac{1}{x^3} \sqrt{\frac{(x^2-1)}{(x^2)}} dx$

9. $\int \sqrt{\frac{x^4}{x^3-1}} dx$

10. $\int \frac{x}{(2x-1)^{\frac{2}{3}}} dx$

Exercises 5.6

Evaluating Definite Integrals

Use Substitution Formula to evaluate the integrals in Exercises 1-3.

1. (a) $\int_0^1 \frac{10\sqrt{v}}{\left(1+v^{\frac{3}{2}}\right)^2} dv,$

(b) $\int_1^4 \frac{10\sqrt{v}}{\left(1+v^{\frac{3}{2}}\right)^2} dv$

2. (a) $\int_{-\frac{\pi}{2}}^0 \left(2 + \tan\frac{t}{2}\right) \sec^2\left(\frac{t}{2}\right) dt$

(b) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(2 + \tan\frac{t}{2}\right) \sec^2\left(\frac{t}{2}\right) dt$

3. $\int_1^4 \frac{dy}{(2\sqrt{y}(1+\sqrt{y})^2)}$

Area

Find the areas of the regions enclosed by the lines and curves in Exercises 4-6.

4. $y = x\sqrt{a^2 - x^2}$, $a > 0$, and $y = 0$.

5. $y = |x^2 - 4|$ and $x = y + 2$.

6. $x - y^{\frac{2}{3}} = 0$ and $x + y^4 = 2$.

Area Between Curves

7. Find the area of the propeller-shaped region enclosed by the curves $x - y^{\frac{1}{3}} = 0$ and $x - y^{\frac{1}{5}} = 0$.

8. Find the area of the region between the curve $y = 3 - x^2$ and the line $y = -1$ by integrating with respect to **(a)** x and **(b)** y .